

Problem 5.3.24

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September 18, 2025

Question: Solve the following pair of linear equations:

$$3x - 5y = 4 \quad (1.1)$$

$$2y + 7 = 9x \quad (1.2)$$

Solution

The equation of a line:

$$\mathbf{n}^T \mathbf{x} = c \quad (2.1)$$

Line L :

$$\begin{pmatrix} 3 & -5 \end{pmatrix} \mathbf{x} = 4 \quad (2.2)$$

Line K :

$$\begin{pmatrix} 9 & -2 \end{pmatrix} \mathbf{x} = 7 \quad (2.3)$$

Combining both the equations:

$$\begin{pmatrix} 3 & -5 \\ 9 & -2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 4 \\ 7 \end{pmatrix} \quad (2.4)$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 9 & -2 & 7 \end{array} \right) \quad (2.5)$$

Solution

Perform the row operation $R_2 \rightarrow R_2 - 3R_1$:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 0 & 13 & -5 \end{array} \right) \quad (2.6)$$

Perform the row operation $R_2 \rightarrow \frac{1}{13}R_2$:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 0 & 1 & -\frac{5}{13} \end{array} \right) \quad (2.7)$$

Back substitute using $R_1 \rightarrow R_1 + 5R_2$:

$$\left(\begin{array}{cc|c} 3 & 0 & \frac{27}{13} \\ 0 & 1 & -\frac{5}{13} \end{array} \right) \quad (2.8)$$

Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} \frac{9}{13} \\ -\frac{5}{13} \end{pmatrix} \quad (2.9)$$

Final Answer: Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} \frac{9}{13} \\ \frac{5}{13} \end{pmatrix}$$

Graph

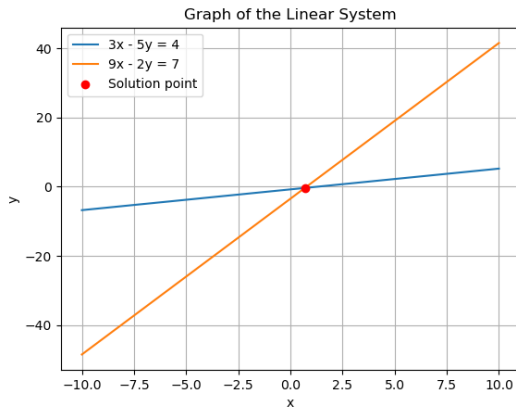


Figure: Vector Representation

```
include "matfun.h"

// Solve 2x2 system using Cramer's rule (appropriate for this
// fixed size)
void solve_linear_system(double A[2][2], double B[2], double X
[2]) {
    double det = A[0][0]*A[1][1] - A[0][1]*A[1][0];
    if (det == 0) {
        // No unique solution
        X[0] = X[1] = 0;
        return;
    }
    X[0] = (B[0]*A[1][1] - B[1]*A[0][1]) / det;
    X[1] = (A[0][0]*B[1] - A[1][0]*B[0]) / det;
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load shared library
lib = ctypes.CDLL('./matfun.so')

# Define C function argument and return types
lib.solve_linear_system.argtypes = [
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double)
]
lib.solve_linear_system.restype = None
```



```
# Define matrix A and vector B for the system
A_c = (ctypes.c_double * 4)(3, -5, 9, -2)
B_c = (ctypes.c_double * 2)(4, 7)
X_c = (ctypes.c_double * 2)()

# Call the C function to solve the system
lib.solve_linear_system(A_c, B_c, X_c)

# Print solution as a column vector
print("Solution (column vector form):")
print(f"\n [{X_c[0]}],\n [{X_c[1]}]\n")
```

```
# Plot the lines corresponding to the equations
a1, b1 = 3, -5
c1 = 4
a2, b2 = 9, -2
c2 = 7

x_vals = np.linspace(-10, 10, 400)
y1 = (c1 - a1 * x_vals) / b1
y2 = (c2 - a2 * x_vals) / b2

plt.plot(x_vals, y1, label='3x - 5y = 4')
plt.plot(x_vals, y2, label='9x - 2y = 7')
```

```
# Plot the solution point
plt.scatter(X_c[0], X_c[1], color='red', label='Solution point',
            zorder=5)

plt.xlabel('x')
plt.ylabel('y')
plt.title('Graph of the Linear System ')
plt.legend()
plt.grid(True)
plt.savefig("fig1.png")
plt.show()
```

```
import numpy as np
import matplotlib.pyplot as plt

# Coefficient matrix and RHS from the image
A = np.array([
    [3, -5],
    [9, -2]
])
b = np.array([4, 7])

# Solve for vector x
x = np.linalg.solve(A, b)
```

```
# Print solution in matrix form
print("Solution vector x:")
print(np.array([[x[0]], [x[1]]]))

# Plot both lines
fig, ax = plt.subplots()

# Line L:  $3x - 5y = 4 \Rightarrow y = (3x - 4)/5$ 
x_vals = np.linspace(-2, 6, 100)
y_L = (3*x_vals - 4)/5
ax.plot(x_vals, y_L, label='3x - 5y = 4 (L)')
```

Python Code

```
# Line K:  $9x - 2y = 7 \Rightarrow y = (9x - 7)/2$ 
y_K = (9*x_vals - 7)/2
ax.plot(x_vals, y_K, label='9x - 2y = 7 (K)')

# Mark intersection point
ax.plot(x[0], x[1], 'ro', label='Intersection')

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.legend()
ax.grid(True)
ax.set_title('Graph of the two lines')

plt.savefig('fig1.png')
plt.show()
```